

Exam 2

- ① SFRs exhibit several inherent safety features. For example, the coolant (typically sodium) is a liquid at a large range of temperatures. This effect is twofold. First, the reactor can operate near atmospheric pressure. Second, there exists a large margin to boiling.

Additionally, although the metallic fuel has a lower melting point than oxide fuel, its thermal conductivity is so much higher that it exhibits a larger margin to melting.

Also, even when there is a cladding failure of metallic fuel, the cladding does not significantly deform as it would in oxide fuels.

- ② Alloying uranium with other bcc metals, like Zirconium, extends the temperature range in which the bcc uranium phase (γ -phase) is stable. Gamma phase U demonstrates much better swelling characteristics than the low-temperature alpha phase. Due to its orthorhombic lattice structure, α -phase uranium swells anisotropically.

- ③ Initial metallic fuel designs did not include a low smear density and found high cladding stresses — or even failed cladding — at relatively low burnups. This is because metallic fuels exhibit significant swelling at low burnups due to swelling fission products and porosity. A low smear

density, typically around 75%, allows the fuel slug to expand into the annular gap before coming into contact with the cladding and causing FCMI. The large plenum is to establish a space for fission product gases to vent to when interconnected porosity is reached, so as to prevent a significant internal pressure that could induce a large stress in the cladding.

- ④ Constituent redistribution in metallic fuels is the process by which Zr diffuses up the temperature gradient, forming three distinct zones in the fuel slug, from center: a Zr-enriched zone, a Zr-depleted zone, and an as fabricated zone. Uranium diffuses in the opposite manner and exists in a depleted manner in the center of the fuel. The phase boundaries vary axially and are determined by temperature. This phenomenon is due to both a temperature gradient (Soret diffusion) and a chemical potential driver between the phases. ^{In U-Zr fuels, a lower} U concentration at the center would result in a lower fission rate in this region and lower centerline temperature.

⑤ Thermal conductivity in metallic fuel degrades with burnup, since porosity increases and the pores have very low thermal conductivity. However, at the point where porosity interconnects and the sodium bond is able to infiltrate the pores, the thermal conductivity rebounds due to the excellent conductivity of sodium.

⑥ Fuel cladding chemical interaction (FCCI) is the interaction/diffusion between chemical species in the fuel and cladding that can occur when the fuel swells. There are two primary adverse effects of FCCI: a thinning and weakening of the cladding that can cause failure and the production of low melting point phases in the fuel. The primary fuel species participating in FCCI are U, Pu, Zr, and lanthanide fission products. Higher temperatures and burnups increase the threat of FCCI because larger amounts of FPs can diffuse to the interface. The main cladding species in FCCI include Fe, Ni, and Cr.

⑦ In MOX fuels at sufficiently high temperatures, four distinct regions form in fuel pellets:

1. a central void where pores coalesce
2. a columnar region with very extended grains, initial microstructure gone
3. equiaxed region with grains that are longer than as-fabricated
4. as-fabricated region

These distinct regions develop due to the movement of pores through the fuel up the temperature gradient. The pores have very low thermal conductivity, so the inner side of the pore is substantially hotter than the outer side. Material evaporates off the inner side and condenses on the colder outer side. The pore therefore moves up the temperature gradient and erases the initial microstructure and forming extended columnar grains. This requires very high temperatures (upwards of 1800°C), so this effect is diminished in the outer region of the fuel pin. Additionally, the composition of the fuel is affected in these regions due to differences in volatility. The more volatile uranium is more likely to evaporate and deposit on the outer side of the pore. This results in a decreased U concentration in the center and a higher concentration of the less volatile Pu, resulting in a higher fission rate and higher temperature.

- ⑧ In addition to the Pu concentration effects from restructuring discussed in the previous question, there can also be regions of high Pu concentration due to manufacturing. These regions experience higher burnups and higher porosity.

The oxygen stoichiometric ratio in the fuel initially is slightly below 2 (1.93-2). Oxygen diffuses down the temperature gradient, so the ratio is closer to 2 in the outer regions of fuel.

- ⑨ Joint Oxide Grain (JOG) is oxide buildup in the gap between the fuel and cladding due to the diffusion of volatile species down the temperature gradient. This oxide acts as a mechanical buffer and reduces fuel swelling, but also plays a large role in initiating FCI.

- ⑩ There are two types of sodium corrosion in SFRs:

(i) the dissolving of alloy constituents in the sodium coolant

(ii) the reaction of alloy constituents with impurities in the sodium

A concern for type (i) is thermal gradient mass transfer whereby soluble elements are deposited on the colder side of the loop due to decreased solubility.